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Asperidine B: Total synthesis and structure correction

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ABSTRACT

A synthesis of the proposed structure of asperidine B is reported, involving a tandem hydroformylation-condensation reaction, a diastereoselective dihydroxylation and the addition of an organozinc reagent to an iminium ion. As the spectroscopic data for the synthetic material is not in agreement with that of the natural product, the structure of asperidine B is reassigned as (+)-preussin.

Introduction

Recently, one of our laboratories reported the isolation of three new alkaloids from the soil-derived fungus *Aspergillus sclerotiorum*. [1] Named asperidines A, B and C, these alkaloids were assigned structures [1–3] respectively (Fig. 1). The structural assignments were on the basis of spectroscopic studies, with that of asperidine C additionally secured by X-ray crystallography.

The other laboratory has been involved in the synthesis of piperidine alkaloids using tandem-hydroformylation condensation [2] to form the heterocyclic ring (Fig. 2) [3]. These efforts have resulted in the syntheses of pseudoconhydrin **4** [4], 5-hydroxysedamine **5** [5] and, most notably in this context, azimic acid **6** [6].

Results and discussion

Given the strong structural similarity between asperidine B **2** and azimic acid **6**, we therefore embarked upon a synthesis of the former alkaloid using the same overall strategy. We initially explored a route to racemic asperidine B (Scheme 1). Treatment of nonanal with allyltrimethylsilane in the presence of toluenesulfonamide and boron trifluoride etherate delivered homoallylic sulfonamide **7** in 74% yield in a tandem Sakurai process [7]. Subjecting this material to the hydroformylation conditions that we had used previously [4–6], specifically with rhodium acetate and biphephos [8] as the ligand under a pressure of 60 psi of a 1:1 hydrogen – carbon monoxide mixture, delivered tetrahydropyridine **8** in 78% yield. Dihydroxylation under Upjohn conditions with added methanesulfonamide delivered the diols **9** as a 3.6:1 mixture in 95% yield. This is a notable lower d.r. than in our previous work [4–6]. Based upon our earlier work, we anticipated that the diols would be unstable. They were, therefore, acetylated. The mixture of diacetates **10** was obtained in 64% yield. At this point we sought a suitable benzylic nucleophile in order to carry out the substitution at the α-position. It has been reported that tribenzylaluminium may be generated from the reaction between benzyl Grignard reagents and aluminium trichloride [9]. The reaction of this mixture with the inseparable diacetates, however, did not lead to the desired product. This is in contrast to our work on azimic acid in which trimethyl aluminium proved to be a viable reagent. We believe that the Grignard derived reagent does not contain free tribenzylaluminium. The attempted use of tetrabenzyltitanium was also fruitless [10]. We did note that the only product obtained from some of these runs was enol acetate ester **11** which was also formed when acetates **10** were allowed to stand at room temperature for an extended period of time. Inspired by the work of Ley *et al.* [11], we employed a mixture of zinc iodide [12] and benzylmagnesium chloride [13] which, we presume, generates a benzylzinc species *in situ*. Reaction of this system [14] with the diacetate mixture delivered a mixture of three isomeric substitution products. The ratio of isomers varied from 3.3:1.3:1 to 2:1.5:1. The

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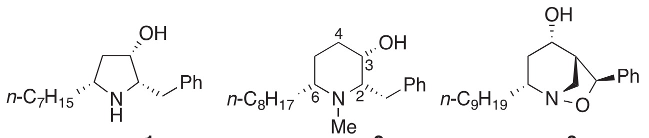
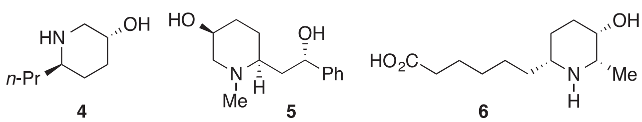


1 2 3

Fig. 1. Alkaloids from *Aspergillus sclerotiorum*.



4 5 6

Fig. 2. Alkaloids synthesised by tandem hydroformylation-condensation.

major product, the all *cis*-isomer **12a**, could be isolated in pure form in 39% yield by flash chromatography [15]. The acetate group was then cleaved using potassium carbonate in methanol to give alcohol **13** in 87% yield. The stereochemistry of this material was subsequently confirmed by X-ray crystallography (Fig. 3) [16]. The tosyl group was removed with sodium naphthalenide to give the secondary amine **14**. Finally, the nitrogen atom was methylated under Eschweiler-Clarke conditions to give piperidine **2**.

The structure of the final product **2** was confirmed by both 1D and 2D NMR spectroscopy, including a NOESY experiment to confirm the all *cis*-stereochemistry. To our concern, the ^{1}H and ^{13}C NMR spectra differed significantly from those reported for the natural product [1]. In particular, there were major differences in the signals for the protons α to the heteroatoms and in the benzylic position. As a result, it was suggested that asperidine B **2** could share a similar pyrrolidine skeleton to asperidine A **1**. As reported in the structural elucidation of **2** [1], the *N*-methylpiperidine moiety was established based on the ^{1}H-^{1}H COSY correlations of H-2/H-3/H$_{ab}$-4/H$_{ab}$-5/H-6 together with HMBC correlations of the *N*-Me with C-2 and C-6 [1]. Due to the overlap of the resonances of H$_a$-4 and H-6 in the 300 MHz ^{1}H NMR spectrum of **2**, this was re-recorded using a 500 MHz NMR spectrometer, resulting in the well-resolved resonances of these protons. Furthermore, the ^{1}H-^{1}H COSY correlations clearly displayed the cross peaks of H-6 with not only H$_{ab}$-5 but also H$_{ab}$-4 and the absence of the connectivity of H$_{ab}$-4/H$_{ab}$-5. Based on these data along with the above HMBC correlations, asperidine B is identified as 2-benzyl-3-hydroxy-5-nonyl-*N*-methylpyrrolidine. The NOESY data of H-2/H-3 and H-5 established their *cis* relationship. The absolute configuration of the *N*-methylpyrrolidine moiety was assigned as 2*S*, 3*S* and 5*R* on the basis of the 3*S* configuration of the secondary alcohol [1] which was determined using the modified Mosher's method [17]. Accordingly, asperidine B has an identical structure to (+)-preussin **15** (Fig. 4) [18]. It is worthwhile to clarify that the specific rotation of asperidine B, $[\alpha]_D^{24}$ + 21 (c 0.6, CHCl$_3$), is similar to that of (+)-preussin, $[\alpha]_D^{24}$ + 19 (c 0.6, CHCl$_3$) [19]. The previous specific rotation was reported to have a negative sign [1], possibly due to an instrument error.

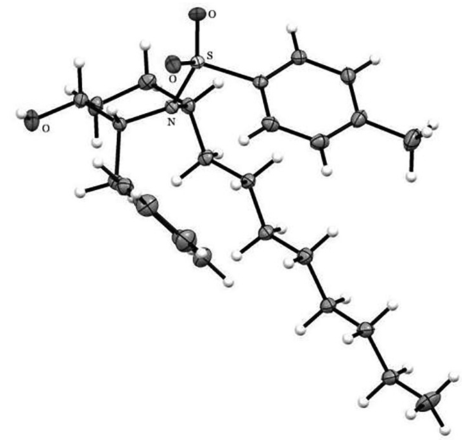


Fig. 3. ORTEP structure of piperidine **13**.

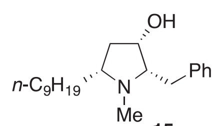


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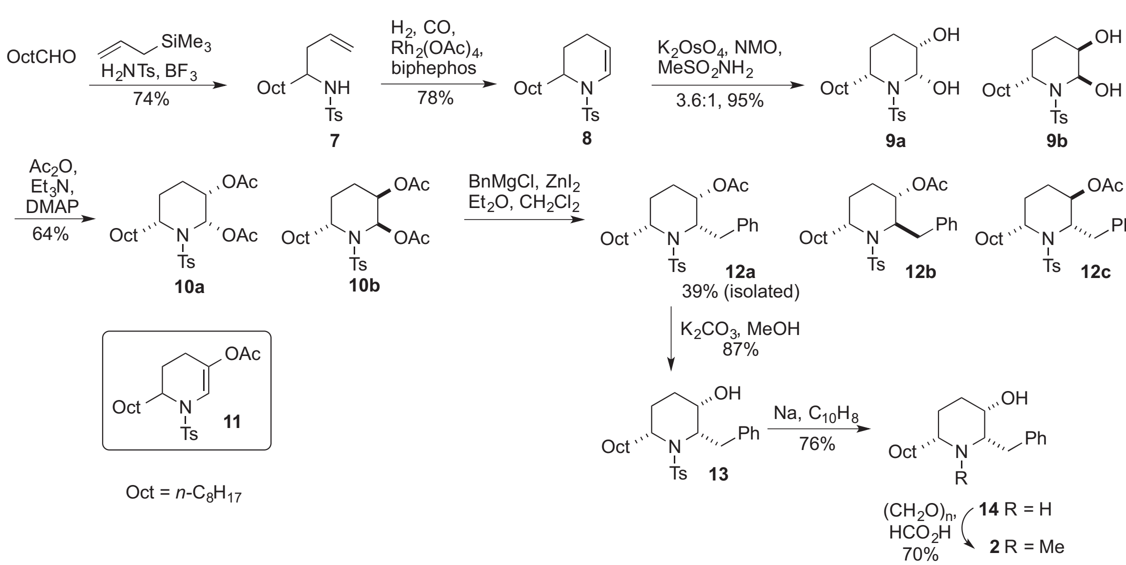
Fig. 4. (+)-preussin.



Scheme 1. Synthesis of the Asperidine B structure.

Conclusion

Total synthesis of the proposed structure of asperidine B has prompted a re-evaluation of the spectroscopic data and a reassignment of the structure. Asperidine B is actually the known alkaloid (+)-preussin. This work highlights the challenges faced by natural products chemists in structure determination and also the contribution that can be made by total synthesis in this endeavor [20].

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tetlet.2020.152078.

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[15] While the acetates were partially separable with some difficulty, the corresponding alcohols were more easily separated.

[16] Details of the X-ray structure determination have been deposited with the Cambridge Crystallographic Data Centre and may be obtained at http://www.ccdc.cam.ac.uk; CCDC deposition number: 1993387. The stereochemistry of the other isomers was determined by spectroscopic studies on the corresponding alcohols.

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